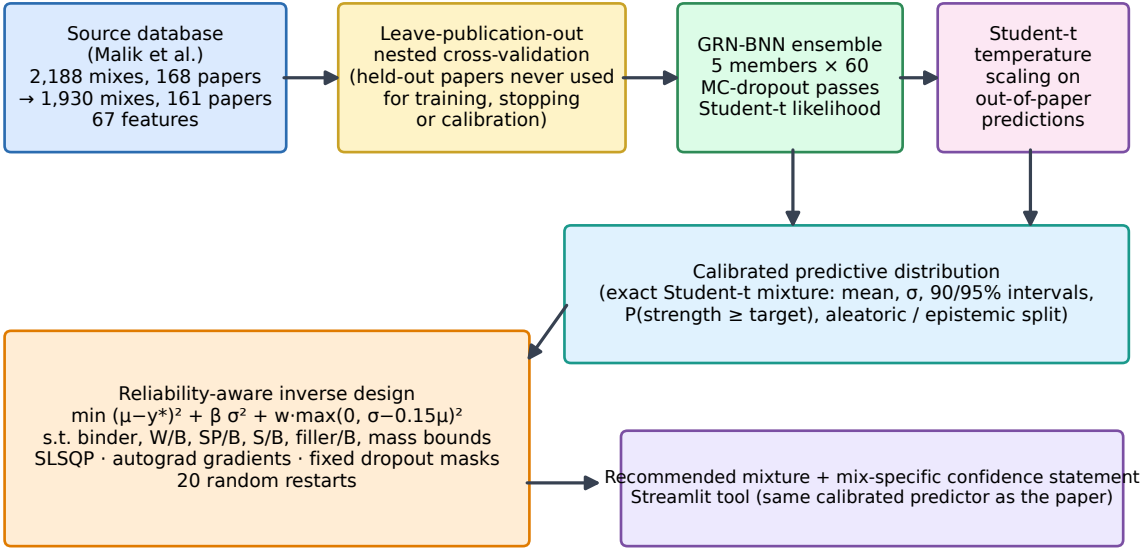
Source database
(Malik et al.)
2,188 mixes, 168 papers
→ 1,930 mixes, 161 papers
67 features
Leave-publication-out
nested cross-validation
(held-out papers never used
for training, stopping
or calibration)
GRN-BNN ensemble
5 members × 60
MC-dropout passes
Student-t likelihood
Student-t
temperature
scaling on
out-of-paper
predictions
Calibrated predictive distribution
(exact Student-t mixture: mean, σ, 90/95% intervals,
P(strength ≥ target), aleatoric / epistemic split)
Reliability-aware inverse design
min (μ−y*)² + β σ² + w·max(0, σ−0.15μ)²
s.t. binder, W/B, SP/B, S/B, filler/B, mass bounds
SLSQP · autograd gradients · fixed dropout masks
20 random restarts
Recommended mixture + mix-specific confidence statement
Streamlit tool (same calibrated predictor as the paper)